

Operations Research for Health Care

Lesson 4/15

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UFMG, Belo Horizonte, Brasil
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Outline of this lecture

Discrete location problems implementation

Set-covering

Maximal covering

P-center

P-median

Fixed-charge

Hierarchical

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Discrete location problems [Daskin and Dean, 2004]

Gnu Linear Programming Kit (GLPK) [Makhorin, 2008]:

- ▶ **Covering-based problems:**

- ▶ Set-covering
- ▶ Maximal covering
- ▶ P-center

- ▶ **Median-based problems:**

- ▶ P-median
- ▶ Fixed-charge
- ▶ Hierarchical

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Set-covering [Ahmadi-Javid et al., 2017]

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

▶ d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.

▶ D_i The maximum acceptable travel time from demand $i \in I$ (the cover time).

$N_i = \{j \in J : d_{ij} \leq D_i\}$: Set of all candidate locations that can cover demand i .

▶ f_j The fixed cost of locating HCU at candidate location $j \in J$.

▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.

$$\min. \sum_{j \in J} f_j x_j \quad (1)$$

$$\text{s.t.: } \sum_{j \in N_i} x_j \geq 1 \quad \forall i \in I \quad (2)$$

$$x_j \in \{0, 1\} \quad \forall j \in J$$

Set-covering [Ahmadi-Javid et al., 2017]

```
set I;
set J;
param d{I,J};
param D{I};
set N{i in I} := setof{j in J: d[i,j] <= D[i]} j;
param f{J};
var x{j in J}, >=0, binary;

minimize OF1: sum{j in J}f[j]*x[j];
s.t. R2{i in I}: sum{j in N[i]}x[j] >= 1;
solve;

printf:"=====";
printf:"Total Cost:  %.2f", OF1;
printf:"Selected Locations:  %d", sum{j in J}x[j];
printf:"=====";
printf{j in J, i in I: x[j]>0}:"[%s] <-- [%s]:  %d km ($ %.2f);", j, i,
d[i,j]*x[j], f[j]*x[j];
printf:"=====";
```

Set-covering [Ahmadi-Javid et al., 2017]

10 origens
5 locais candidatos

Matriz de distâncias (km)

	1	2	3	4	5	
1	98	50	93	90	70	(79)
2	86	52	84	84	86	(75)
3	62	31	47	97	68	(75)
4	32	52	33	91	31	(48)
5	72	47	66	95	34	(76)
6	77	37	54	46	99	(79)
7	59	46	99	91	80	(75)
8	54	95	93	59	92	(75)
9	47	42	95	39	43	(79)
10	55	54	91	90	33	(75)

(valores: 110 120 90 90 85
#203
(5))

Set-covering [Ahmadi-Javid et al., 2017]

```

set I;
set J;
param d{I,J};
param D{I};
set N{i in I} := setof{j in J: d[i,j] <= D[i]} j;
param f{J};
var x{j in J}, >=0, binary;

minimize OF1: sum{j in J} f[j]*x[j];
s.t. R2{i in I}: sum{j in N[i]} x[j] >= 1;
solve;

printf: "=====";
printf: "Total Cost: %.2f", OF1;
printf: "Selected Locations: %d", sum{j in J} x[j];
printf: "=====";
printf{j in J, i in I: x[j]>0}: "[%e] <-- [%e]: %d km ($ %.2f)", j, i,
d[i,j]*x[j], f[j]*x[j];
printf: "=====";
    
```



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Set-covering

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Maximal covering [Ahmadi-Javid et al., 2017]

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

- ▶ d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.
- ▶ D_i The maximum acceptable travel time from demand $i \in I$ (the cover time).

$N_i = \{j \in J : d_{ij} \leq D_i\}$: Set of all candidate locations that can cover demand i .

- ▶ w_i The demand point at $i \in I$.
- ▶ p The number of candidate locations to be established.
- ▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.
- ▶ $z_i \in \{0, 1\}$ 1, if demand point $i \in I$ is covered; 0 otherwise.

Maximal covering [Ahmadi-Javid et al., 2017]

$$\max. \sum_{i \in I} w_i z_i \quad (4)$$

$$\text{s.t.: } \sum_{j \in J} x_j = p \quad (5)$$

$$z_i \leq \sum_{j \in N_i} x_j \quad \forall i \in I \quad (6)$$

$$x_j \in \{0, 1\} \quad \forall j \in J$$

$$z_i \in \{0, 1\} \quad \forall i \in I$$

Maximal covering [Ahmadi-Javid et al., 2017]

```
set I;  
set J;  
param d{I,J};  
param D{I};  
set N{i in I} := setof{j in J: d[i,j] <= D[i]} j;  
param w{I};  
param p;  
var x{j in J}, >=0, binary;  
var z{i in I}, >=0, binary;  
  
maximize OF4:  sum{i in I}w[i]*z[i];  
s.t.  R5:  sum{j in J}x[j] = p;  
s.t.  R6{i in I}:  z[i] <= sum{j in N[i]}x[j];
```

Maximal covering [Ahmadi-Javid et al., 2017]

10 origins
5 locos candidatos

Matriz Distâncias Demanda (mex)

	1	2	3	4	5	
1	99	50	93	90	70	36 (79)
2	86	52	84	84	86	51 (50) ←
3	62	31	47	87	68	46 (75)
4	32	52	33	91	31	41 (78)
5	72	47	66	95	34	50 (76)
6	77	37	54	46	98	58 (79)
7	57	46	99	91	50	51 (75)
8	34	95	93	58	92	44 (45) ←
9	47	42	95	39	43	59 (77)
10	55	54	91	90	33	30 (75)

(valos: 110 120 90 90 85

7203

(D) Seleção de 2 (origens limitadas)

Maximal covering [Ahmadi-Javid et al., 2017]

```

set I;
set J;
param d[I,J];
param D[I];
set N[i in I] := setof{j in J: d[i,j] <= D[i]} j;
param w[I];
param p;
var x{j in J}, >=0, binary;
var z{i in I}, >=0, binary;

maximize OF4: sum{i in I} w[i]*z[i];
s.t. R5: sum{j in J} x[j] = p;
s.t. R6{i in I}: z[i] <= sum{j in N[i]} x[j];
    
```



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Maximal covering

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Hierarchical

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

► d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.

► D_i The maximum acceptable travel time from demand $i \in I$ (the cover time).

$N_i = \{j \in J : d_{ij} \leq D_i\}$: Set of all candidate locations that can cover demand i .

► w_i The demand point at $i \in I$.

► p The number of candidate locations to be established.

► $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.

► $y_{ij} \in \{0, 1\}$ 1, if demand point i is assigned to a facility at candidate location $j \in N_i$; 0 otherwise.

► $L \geq 0$ L is an auxiliary variable (not a decision variable) that is used to compute the maximum distance.

$$\min. L \quad (9)$$

$$\text{s.t.: } \sum_{j \in N_i} y_{ij} = 1 \quad \forall i \in I \quad (10)$$

$$\sum_{j \in J} x_j = p \quad (11)$$

$$\sum_{j \in N_i} w_i d_{ij} y_{ij} \leq L \quad \forall i \in I \quad (12)$$

$$y_{ij} \leq x_j \quad \forall i \in I, j \in N_i \quad (13)$$

$$x_j \in \{0, 1\} \quad \forall j \in J$$

$$y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$$


```
set I;
set J;
param d{I,J};
param D{I};
set N{i in I} := setof{j in J: d[i,j] <= D[i]} j;
param w{I};
param p;
var x{j in J}, >=0, binary;
var y{i in I, j in N[i]}, >=0, binary;
var L, >= 0;

minimize OF9: L;
s.t.  R10{i in I}:  sum{j in N[i]}y[i,j] = 1;
s.t.  R11:  sum{j in J}x[j] = p;
s.t.  R12{i in I}:  sum{j in N[i]}w[i]*d[i,j]*y[i,j] <= L;
s.t.  R13{i in I, j in N[i]}:  y[i,j] <= x[j];
```

10 origens
5 locais candidatos

Matriz Distâncias

	1	2	3	4	5		(max)
1	93	50	93	90	70	34	(79)
2	86	52	84	94	86	51	(78)
3	62	31	47	97	68	46	(75)
4	32	52	33	91	31	41	(78)
5	32	47	66	95	34	50	(76)
6	77	37	54	46	94	54	(79)
7	57	46	98	91	50	51	(78)
8	54	95	93	59	92	44	(76)
9	47	42	95	39	43	59	(77)
10	55	54	91	90	33	36	(75)

(valos: 110 120 90 90 85

1200

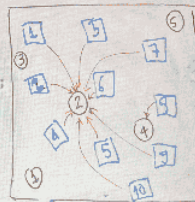
(5) Seleção de 2 (diagonal limitado)

P-center [Ahmadi-Javid et al., 2017]

```

set I;
set J;
param d{I,J};
param D{I};
set N{i in I} := setof{j in J: d[i,j] <= D[i]} j;
param w{I};
param p;
var x{j in J}, >=0, binary;
var y{i in I, j in N[i]}, >=0, binary;
var L, >= 0;

minimize OF9: L;
s.t. R10{i in I}: sum{j in N[i]} y[i,j] = 1;
s.t. R11: sum{j in J} x[j] = p;
s.t. R12{i in I}: sum{j in N[i]} w[i]*d[i,j]*y[i,j] <= L;
s.t. R13{i in I, j in N[i]}: y[i,j] <= x[j];
    
```



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P-center

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Maximal covering

P-center

P-median

Fixed-charge

Hierarchical

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

- ▶ d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.
- ▶ w_i The demand point at $i \in I$.
- ▶ p The number of candidate locations to be established.
- ▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.
- ▶ $y_{ij} \in \{0, 1\}$ 1, if demand point i is assigned to a facility at candidate location $j \in N_i$; 0 otherwise.

$$\min. \sum_{i \in I} \sum_{j \in J} w_i d_{ij} y_{ij} \quad (17)$$

$$\text{s.t.: } \sum_{j \in J} y_{ij} = 1 \quad \forall i \in I \quad (18)$$

$$\sum_{j \in J} x_j = p \quad (19)$$

$$y_{ij} \leq x_j \quad \forall i \in I, j \in N_i \quad (20)$$

$$x_j \in \{0, 1\} \quad \forall j \in J$$

$$y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$$

```
set I;
set J;
param d{I,J};
param w{I};
param p;
var x{j in J}, >=0, binary;
var y{i in I, j in J}, >=0, binary;

minimize OF17: sum{i in I, j in J}w[i]*d[i,j]*y[i,j];
s.t.  R18{i in I}: sum{j in J}y[i,j] = 1;
s.t.  R19: sum{j in J}x[j] = p;
s.t.  R20{i in I, j in J}: y[i,j] <= x[j];
```

P-median [Ahmadi-Javid et al., 2017]

10 origens
5 locais candidatos

Matriz Distâncias

	1	2	3	4	5	Distância (m)
1	98	50	93	90	70	(79)
2	86	52	84	84	86	(75)
3	62	31	47	17	68	(75)
4	32	52	33	91	31	(78)
5	72	47	66	95	34	(76)
6	77	37	54	46	99	(79)
7	52	46	99	91	50	(75)
8	34	95	93	59	92	(75)
9	47	42	95	38	43	(77)
10	55	54	91	90	33	(75)

(valos: 110 120 90 90 85
fpos
(f) Seleção de $\geq$ (candidato limitado)

P-median [Ahmadi-Javid et al., 2017]

```

set I;
set J;
param d{I,J};
param w{I};
param p;
var x{j in J}, >=0, binary;
var y{i in I, j in J}, >=0, binary;

minimize OF17: sum{i in I, j in J} w[i]*d[i,j]*y[i,j];
s.t. R18{i in I}: sum{j in J} y[i,j] = 1;
s.t. R19: sum{j in J} x[j] = p;
s.t. R20{i in I, j in J}: y[i,j] <= x[j];
    
```



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P-median

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P-center

P-median

Fixed-charge

Hierarchical

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

- ▶ d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.
- ▶ f_j Fixed charge of establishing a facility at candidate location $j \in J$.
- ▶ v Variable transportation cost per item per distance unit.
- ▶ U_j facility Capacity (In a capacitated Fixed-charge Location Problem).
- ▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.
- ▶ $y_{ij} \in \{0, 1\}$ 1, if demand point i is assigned to a facility at candidate location $j \in N_i$; 0 otherwise.

$$\min. \sum_{j \in J} f_j x_j + v \sum_{i \in I} w_i d_{ij} y_{ij} \quad (23)$$

$$\text{s.t.: } \sum_{j \in J} y_{ij} = 1 \quad \forall i \in I \quad (24)$$

$$y_{ij} \leq x_j \quad \forall i \in I, j \in N_i \quad (25)$$

$$\sum_{i \in I} w_i y_{ij} \leq U_j \quad \forall j \in J \quad (26)$$

$$x_j \in \{0, 1\} \quad \forall j \in J$$

$$y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$$

```
set I;
set J;
param d{I,J};
param w{I};
param f{J};
param v;
param U{J};
var x{j in J}, >=0, binary;
var y{i in I, j in J}, >=0, binary;

minimize OF23: sum{j in J}f[j]*x[j] + v*sum{i in I, j in
J}w[i]*d[i,j]*y[i,j];
s.t. R24{i in I}: sum{j in J}y[i,j] = 1;
s.t. R25{i in I, j in J}: y[i,j] <= x[j];
s.t. R26{j in J}: sum{i in I} w[i]*y[i,j] <= U[j];
```

Fixed-charge [Ahmadi-Javid et al., 2017]

20 prisms	
5 locusts candidates	
Matry Distancias	Jamardo (m)
1	98 50 93 90 70 36 (79)
2	86 52 84 84 86 52 (75)
3	62 31 47 87 69 46 (75)
4	32 52 33 91 31 41 (72)
5	72 47 66 95 34 50 (76)
6	77 37 51 46 98 54 (70)
7	58 46 98 91 50 51 (75)
8	34 75 33 58 92 44 (75)
9	47 42 95 38 43 59 (73)
10	55 54 91 50 33 30 (75)

(F) 110 120 90 90 85
CAP: 195 220 210 105 100
CV (b/w): 5

Fixed-charge [Ahmadi-Javid et al., 2017]

```

set I;
set J;
param d{I,J};
param w{I};
param f{J};
param v;
param U{J};
var x{j in J}, >=0, binary;
var y{i in I, j in J}, >=0, binary;

minimize OF23: sum{j in J} f[j]*x[j] + v*sum{i in I,
J} w[i]*d[i,j]*y[i,j];

s.t. R24{i in I}: sum{j in J} y[i,j] = 1;
s.t. R25{i in I, j in J}: y[i,j] <= x[j];
s.t. R26{j in J}: sum{i in I} w[i]*y[i,j] <= U[j];

```



Outline of this lecture

Discrete location problems implementation

Set-covering

Maximal covering

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P-median

Fixed-charge

Hierarchical

I : The set of demand points.

K : The set of candidate locations for a level-1 PCF (e.g., clinics).

J : The set of candidate locations for a level-2 PCF (e.g., hospitals).

- ▶ $d1_{ik}$ The travel time from demand point $i \in I$ to a level-1 PCF in a candidate location $k \in K$.
- ▶ $d2_{kj}$ The travel time from a level-1 PCF in a candidate location $k \in K$ and a level-2 PCF in a candidate location $j \in J$.
- ▶ w_i The population size, demand point at $i \in I$.
- ▶ p The number of a level-1 PCF locations to be established.
- ▶ q The number of a level-2 PCF locations to be established.
- ▶ $C1_k$ The capacity of a level-1 PCF in candidate location $k \in K$.
- ▶ $C2_j$ The capacity of a level-2 PCF in candidate location $j \in J$
- ▶ O_k Proportion of level-1 PCF patients at candidate location $k \in K$ referred to a level-2 PCF.

- ▶ $x1_k \in \{0, 1\}$ 1, if a level-1 PCF is established at candidate location $k \in K$; 0 otherwise.
- ▶ $x2_j \in \{0, 1\}$ 1, if a level-2 PCF is established at candidate location $j \in J$; 0 otherwise.
- ▶ u_{ik} The flow of patients between demand point $i \in I$ and a level-1 PCF at candidate location $k \in K$
- ▶ v_{kj} The flow of patients between demand point $k \in K$ and a level-2 PCF at candidate location $j \in J$

$$\min. \sum_{i \in I} \sum_{k \in K} d1_{ik} u_{ik} + \sum_{k \in K} \sum_{j \in J} d2_{kj} v_{kj} \quad (28)$$

$$\text{s.t.} \sum_{k \in K} u_{ik} = w_i \quad \forall i \in I \quad (29)$$

$$\sum_{j \in J} v_{kj} = O_k \sum_{i \in I} u_{ik} \quad \forall k \in K \quad (30)$$

$$\sum_{i \in I} u_{ik} \leq C1_k x1_k \quad \forall k \in K \quad (31)$$

$$\sum_{k \in K} v_{kj} \leq C2_j x2_j \quad \forall j \in J \quad (32)$$

$$\sum_{k \in K} x1_k = p \quad (33)$$

$$\sum_{j \in J} x2_j = q \quad (34)$$

$$x1_k \in \{0, 1\} \quad \forall k \in K$$

$$x2_j \in \{0, 1\} \quad \forall k \in K$$

$$u_{ik} \geq 0 \quad \forall i \in I, k \in K$$

$$v_{kj} \geq 0 \quad \forall k \in K, j \in J$$

```
set I;  
set K;  
set J;  
param d1{I,K};  
param d2{K,J};  
param w{I};  
param p;  
param q;  
param C1{K};  
param C2{J};  
param O{K};  
var x1{k in K}, >=0, binary;  
var x2{j in J}, >=0, binary;  
var u{i in I, k in K}, >=0;  
var v{k in K, j in J}, >=0;
```

```
minimize OF28:  sum{i in I, k in K}d1[i,k]*u[i,k] + sum{k in K, j in
J}d2[k,j]*v[k,j];
s.t.   R29{i in I}:  sum{k in K}u[i,k] = w[i];
s.t.   R30{k in K}:  sum{j in J}v[k,j] = O[k]*sum{i in I}u[i,k];
s.t.   R31{k in K}:  sum{i in I}u[i,k] <= C1[k]*x1[k];
s.t.   R32{j in J}:  sum{k in K}v[k,j] <= C2[j]*x2[j];
s.t.   R33:  sum{k in K}x1[k] = p;
s.t.   R34:  sum{j in J}x2[j] = q;
```

Hierarchical [Ahmadi-Javid et al., 2017]

903202

10 triângulos

5 exatos candidatos

Matriz Distâncias (distância mínima)

	1	2	3	4	5	Max (4)
1	99	50	93	90	70	(79)
2	86	52	84	84	86	(75)
3	62	31	47	17	68	(45)
4	32	52	33	11	31	(48)
5	72	47	65	95	34	(46)
6	77	37	54	46	98	(49)
7	53	46	93	93	50	(75)
8	54	95	93	54	92	(75)
9	47	42	95	38	43	(79)
10	55	54	91	30	33	(75)

(6) = (9)

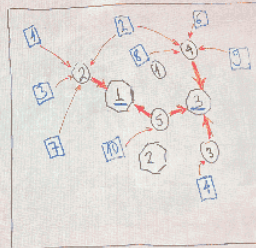
Matriz de Similaridade: 0.8 0.7 0.6 0.5 0.8

CDP: 170 180 30 110 140

Hierarchical [Ahmadi-Javid et al., 2017]

```

set I;
set K;
set J;
param d1{I,K};
param d2{K,J};
param w{I};
param p;
param q;
param C1{K};
param C2{J};
param O{K};
var x1{k in K}, >=0, binary;
var x2{j in J}, >=0, binary;
var u{i in I, k in K}, >=0;
var v{k in K, j in J}, >=0;
    
```



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Hierarchical

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Matriz Dist. (a) - (b)

	1	2	3	Max (2)
1	43	67	45	
2	57	91	80	
3	38	71	55	
4	46	50	30	
5	44	54	55	

CDP: 140 190 200

References

-  Ahmadi-Javid, A., Seyed, P., and Syam, S. S. (2017).
A survey of healthcare facility location.
Computers & Operations Research, 79:223–263.
-  Daskin, M. S. and Dean, L. K. (2004).
Location of health care facilities.
Operations research and health care: A handbook of methods and applications, pages 43–76.
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